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A molecular dynamic study on liquid droplet evaporation under low atmospheric pressure conditions

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A B S T R A C T

Evaporation at high altitudes under low atmospheric pressure has garnered significant attention due to its distinct behavior compared to standard pressures. To address the effects of atmospheric pressure, molecular dynamics simulations are conducted on liquid droplets under three different conditions: 0.1 bar, 0.5 bar, and 1 bar. The temporal evolution of macroscopic parameters and the spatiotemporal dynamics of the liquid droplet are analyzed. The results show that reduced interactions, due to the low number density of nitrogen particles, lead to lower heat absorption by the liquid and a thinner liquid-gas interface, resulting in a lower evaporation rate at low atmospheric pressure. An increased initial evaporation rate at 0.1 bar is observed, resembling evaporation into a vacuum. The discrepancy between the D^2 law and the MD results increases as the vacuum degree rises, suggesting that the D^2 law is not suitable for predicting droplet evaporation behavior under low atmospheric pressure conditions. This work provides a fundamental reference for the design of evaporative cooling systems in high-altitude, low-atmospheric-pressure environments.

1. Introduction

Evaporation occurring at high altitudes under low atmospheric pressures differs significantly from that at standard pressures (1 bar). Experimental results have revealed intriguing phenomena, including reduced mass transfer efficiency [1,2] and diminished total heat transfer performance on the air side [3]. The presence of air molecules with varying number densities influences evaporation behavior through different interaction strengths. Therefore, understanding the mechanisms by which low atmospheric pressure affects evaporation is essential.

The influence of atmospheric pressure on evaporation is ***not*** sufficiently addressed in equilibrium-based descriptions. The evaporation process can be typically further simplified into a convection and diffusion process between saturation vapor around liquid phase and vapor in the air [4–6]. The broadly used saturation vapor pressure is provided by a conceptual framework for phase-transition equilibrium [7]. However, the evidence consistently shows that the actual pressure around the liquid phase within the Knudsen layer is significantly lower than the saturation vapor pressure [8–10]. This suggests that the saturation vapor pressure under the equilibrium descriptions may be inadequate [11]. This discrepancy arises because the equilibrium framework is intrinsically incompatible with the nonequilibrium and subcontinuum nature of the Knudsen layer [12]. Expressions of saturation vapor pressure are typically represented as single-valued functions of temperature [13,14], rather than functions of atmospheric pressure.

Non-equilibrium behaviors should be considered when incorporating the atmospheric pressure effects into the evaporation process. Evaporation primarily occurs a region on the scale of the molecular mean free path (MFP) [15]. The substance under such MFP scales is not a continuous medium, which the application of continuum approaches are constrained. Its dynamics at this scale can be effectively described by the molecular dynamics simulations (MD) [15,16]. Unlike other methods [17], the MD approach does not require prior knowledge of the exact liquid-vapor layer thickness and has consistently demonstrated its effectiveness as a powerful tool for understanding evaporation and condensation processes [18–20]. This highlights the advantages of the MD approach for investigating atmospheric pressure effects.

The effects of atmospheric pressure are studied through MD using both implicit and explicit methods. The implicit method does not introduce additional air molecules but instead employs barostatting techniques [21,22]. The physical parameters of asphalt are studied under 1 atm and the predicted densities were verified by laboratory data [23]. The mechanical properties of processed surface for silica glass are investigated under room temperature and pressures [24]. In contrast, the explicit method directly introduces representative air molecules to model ambient pressures. The explicit method benefits from direct observations of the interactions between representative air molecules and

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the studied substances. The effects of high air pressure (50 bar, 100 bar, 150 bar) induced by nitrogen and oxygen molecules are studied, along with their impact on water evaporation from a silica surface [25]. The effects of low atmospheric pressure on the effective thermal conductivity of a heat pipe were investigated, with results showing that the maximum conductivity is inversely proportional to air pressure [26]. The effects of atmospheric pressure on various physical processes are well studied; however, the impact of low atmospheric pressure on non-equilibrium evaporation dynamics remains an open question.

In this study, we present a non-stationary, non-equilibrium molecular dynamics (NEMD) approach to investigate the effects of atmospheric pressure on liquid argon evaporation. The temporal evolution of macroscopic parameters and the spatial-temporal dynamics of liquid argon are investigated. Firstly, snapshots of the trajectories and evaporation rates are examined to understand the overall evaporation behavior of the argon droplet. Secondly, the radial density profiles are analyzed to explore the temporal evolution of the liquid-vapor interface during evaporation. Thirdly, the temporal temperature of argon is investigated to examine the interactions between argon and nitrogen. Finally, the MD results are compared with the traditional D^2 law, and the discrepancies across three different atmospheric conditions are discussed.

2. Methodology

2.1. Problem statement and simulation configuration

An argon droplet is placed surrounding with nitrogen environment to investigate its evaporation behaviors [27–29]. The number density of nitrogen particles is determined based on the prevailing different atmospheric pressure conditions [30]. The argon particles are expected to absorb heat from the nitrogen molecules at high temperatures to evaporate.

As illustrated in Fig. 1, the size of the simulation domain is $L_x = 40\ nm, L_y = 40\ nm, L_z = 40\ nm$, and periodic boundaries are applied in each direction. The size of the simulation box is selected larger than the liquid argon diameter to avoid additional impacts on the evaporation [31]. A 15 nm diameter liquid argon droplet, containing 34,985 particles at 100 K [32], is placed at the center of the simulation box. The density of surrounding nitrogen molecules varies with pressure: 3.437 kg/m^3 at 1 bar, 1.701 kg/m^3 at 0.5 bar, and 0.33376 kg/m^3 at 0.1 bar. To maintain these specific atmospheric pressures around the liquid argon, the number of nitrogen molecules is kept constant at 453, 2275, and 4597, respectively, throughout the simulation. The same three argon deletion zones, each with a diameter of 30 nm and centered at the geometric center of the simulation box, are defined and applied every 100 timesteps outside the liquid droplet to simulate its evaporation into an infinite space [31,33].

2.2. Simulation configuration

The liquid argon evaporation processes under three different atmospheric pressures are simulated using LAMMPS software [34]. The interactions between particles are described by the 12-6 Lennard-Jones potential,

$$U(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} - \left(\frac{\sigma}{r_c}\right)^{12} + \left(\frac{\sigma}{r_c}\right)^{12}\right], \quad (1)$$

where r is the distance between particles, r_c is the cut-off radius, σ is the particle diameter, and ϵ is the depth of the potential wall. The LJ parameters of particles [35,36] are: $\epsilon_{N_2} = 0.0032\ eV$, $\sigma_{N_2} = 3.3078$ Å, $\epsilon_{Ar} = 0.0103\ eV$, $\sigma_{Ar} = 3.410$ Å.The bond length of nitrogen is fixed at 1.10 Å [37] using SHAKE algorithm [38]. The interactions between argon and nitrogen particles are determined from the Lorentz-Berthelot mixing rule [39].

Three main periods are defined in this simulation: the velocity-scaling period, the equilibration period, and the evaporation period. During the velocity-scaling period, a minimization operation is performed to eliminate particle overlaps in the argon droplet during its initialization. In the equilibration period, all particles are allowed to reach thermodynamic equilibrium under the Nose–Hoover NVT ensemble at 100 K [40]. Equilibration is carried out for 2000 ps with a timestep of 5 fs. Once thermodynamic equilibrium is achieved, the temperature of the nitrogen molecules is abruptly increased to 120 K using the Langevin thermostat [41]. With a 20 K temperature difference, the liquid argon droplet absorbs heat from the surrounding nitrogen, initiating the evaporation process. The evaporation period is set to 1500 ps also with the timestep of 5 fs. The simulation results are visualized using OVITO software in this paper [42].

3. Results and discussion

3.1. Evaporation behaviors

To gain a better understanding of the non-equilibrium evaporation behaviors, the system's trajectory at four different time points is shown in Fig. 2. It should be noted that, after 2000 ps of equilibration, the time step is reset to 0 for the convenience of evaporation analysis. In all snapshots, red particles represent nitrogen molecules, while argon

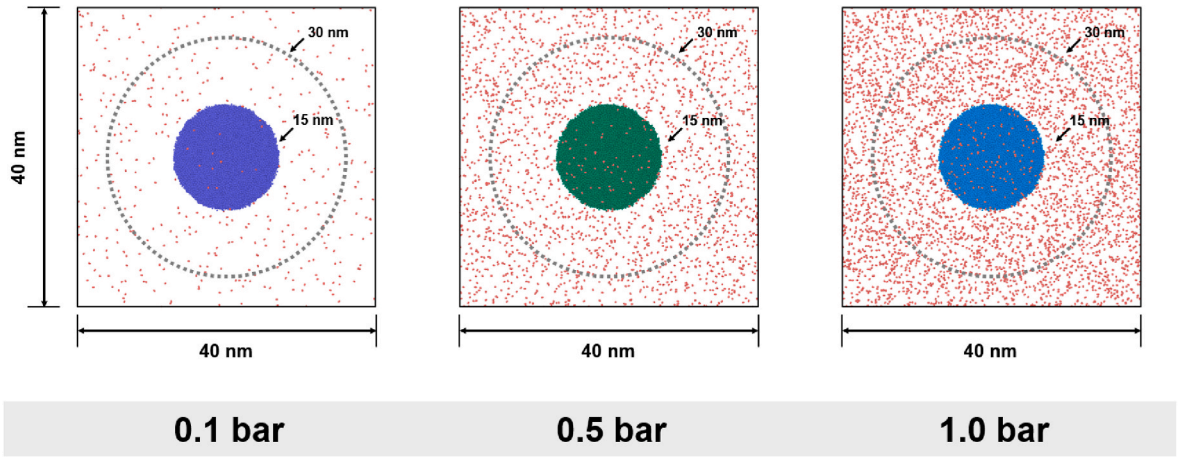


Fig. 1. Initial configurations under three different atmospheric conditions. The argon droplet is placed at the center of the simulation box, with different colors representing different conditions. The nitrogen particles are shown in red, and the grey dotted circle indicates the argon deletion zone.

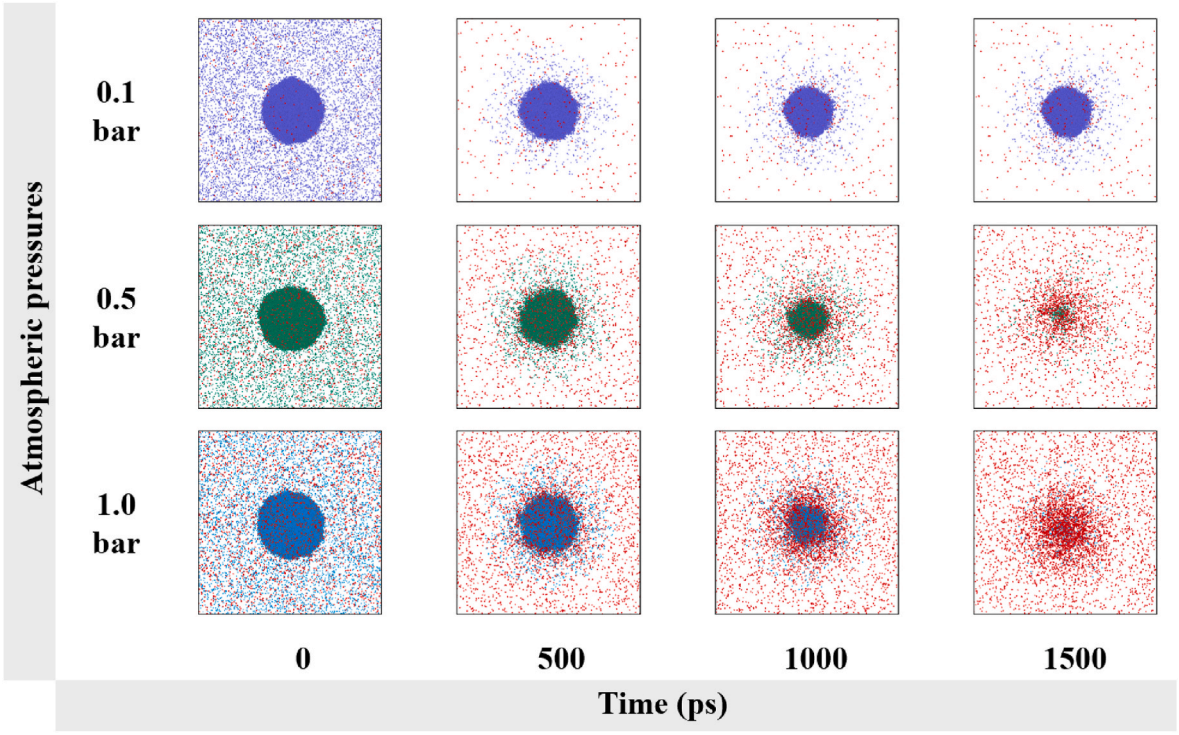


Fig. 2. Snapshots of the liquid argon evaporation behaviors under different atmospheric pressure. The red particles represent nitrogen, while the argon particles are colored differently to better illustrate their behaviors under varying pressures.

particles are colored in three distinct colors for easier comparison.

Snapshots of argon evaporation under different conditions at four distinct times are shown. Since the droplets have the same initial diameter, their evaporation behaviors under various atmospheric conditions can be compared. As the evaporation progresses, the droplet diameter generally decreases. Throughout the simulation, the liquid argon droplet maintains a rough spherical shape, and radial analysis will be conducted later. At the end of the evaporation process, at 1500 ps, the droplet at 1.0 bar appears to have disappeared, while the other two droplets retain their size.

The interactions between argon and nitrogen increase as the collision frequency increases. The interaction can be roughly assessed by the extent of nitrogen aggregation around the liquid argon droplet. Initially, at 0 ps, nitrogen and argon are uniformly distributed due to equilibrium conditions. By 1000 ps at 1.0 bar, a significantly larger amount of nitrogen (shown in red) accumulates around the droplet compared to the other two conditions, indicating a stronger interaction. As a result, the droplet evaporates more rapidly at 1.0 bar than under the other two conditions.

The evaporation rate is directly defined in this study by the number of argon particles removed from the deletion zone. The deletion zone where argon particles are removed lies outside a spherical space with a 30 nm diameter, centered at the mass center of the argon droplet, as indicated by the dotted line in Fig. 1. The evaporation rate can be written as: $J = dN_E/dt$, where N_E is the number of the deleted number. The temporal evaporation rates of the three droplets are shown in Fig. 3. It should be noted that, since the number of nitrogen particles varies, the number of argon particles within a 30 nm diameter also changes after the equilibration period. To account for this effect, the evaporation rate between the first- and second-time step is excluded in the following figure.

At the onset of evaporation, a higher initial evaporation rate of 20.63 num/ps is observed under the 0.1 bar condition. As the evaporation process progresses, the maximum evaporation rates reach 39.96 num/ps at 1.0 bar and 34.36 num/ps at 0.5 bar. However, under 0.1 bar, the

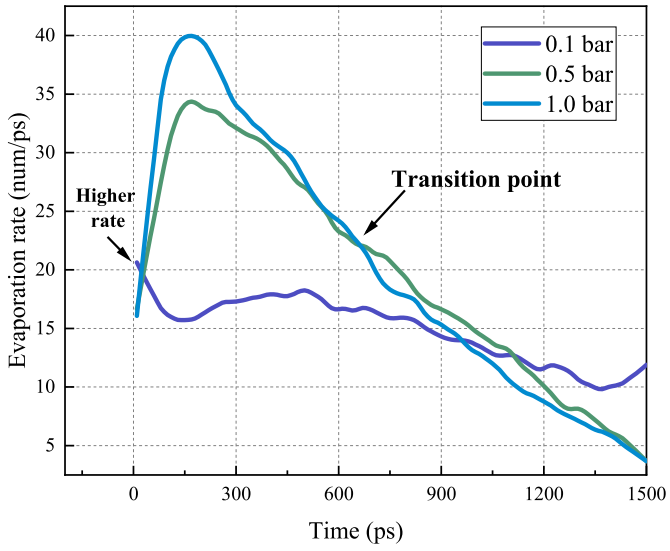


Fig. 3. Temporal evaporation rates under different atmospheric pressures.

evaporation rate generally decreases, reaching a minimum of 15.71 num/ps before gradually increasing. A transition point occurs at 670 ps, where the evaporation rate under 0.5 bar exceeds that under 1.0 bar.

Evaporation behavior under 0.1 bar significantly differs from other conditions: a higher evaporation rate during the initial stage, followed by a decrease and then a subsequent increase in the later stages. These distinct differences can be explained by the evaporation mechanism [15]. Initially, argon particles with high kinetic energy at the interfaces escape quickly, being less hindered by nitrogen molecules due to the low nitrogen density. As evaporation continues, the liquid droplet temperature decreases, resulting in lower thermal motion, reduced fluctuations, and a lower probability of evaporation. Consequently, the

evaporation rate generally decreases. These results are consistent with previous studies [43], particularly the similarity to evaporation into a vacuum [44]. However, in the presence of consistently heated nitrogen particles, which add heat to the droplet, the evaporation rate increases, eventually transitioning into a stable evaporation period. With fewer interactions due to the limited number of nitrogen molecules, the evaporation rate in the latter stable period is significantly lower than under the other two conditions.

Evaporation behaviors under 0.5 bar and 1.0 bar conditions are similar. A transition point around 670 ps is observed, where the evaporation rate alternates between high and low values. Similar results have also been reported in Refs. [31,45], although their observations were made under different ambient temperatures. In our simulations, the droplets are heated under a constant ambient temperature of 120 K. These differences suggest that the heat acquired by argon due to interactions with varying numbers of nitrogen molecules differs. These varying interactions can be considered equivalent to different ambient temperatures with the same number of air molecules. As shown in Fig. 1, the trajectories indicate that the interactions between argon and nitrogen at 1.0 bar are much more intense than at 0.5 bar. This stronger interaction could lead to a higher equivalent ambient air temperature, resulting in more heat being acquired by the argon. In section 3.3, we further analyzed the temperature of the argon to have further discussion on the heat argon absorbed.

3.2. Temporal radial density distribution

The radial density distributions of the liquid droplet are analyzed under three different atmospheric pressures. The center of the droplet is taken as the center of the simulation box, and the radial distance is divided into 4 Å intervals to illustrate the density profile. The distributions at 300 ps, 600 ps, 900 ps, 1200 ps, and 1500 ps are shown in Fig. 4. The thickness of the interface of argon can be identified as the range between $0.95\rho_l$ and $\rho_v + 0.01\rho_l$, where and are the densities of the bulk liquid and vapor, respectively [46].

The thickness of the liquid-vapor interface increases as the ambient atmospheric pressure rises. As shown in Fig. 4, at 300 ps, the thickness of the liquid-gas region is 20 Å, 28 Å, and 32 Å under 0.1 bar, 0.5 bar, and 1.0 bar, respectively. The thickness of this interface remains relatively constant throughout the evaporation process. By 1500 ps, the argon density distribution can be considered to be in the vapor phase, indicating that the droplet has fully evaporated, as also confirmed by the trajectory figure.

The thickness of the liquid-vapor interface is closely related to evaporated mass flux [33,47] and droplet temperature [48]. The evaporative mass flux is significantly increased in the interfacial region,

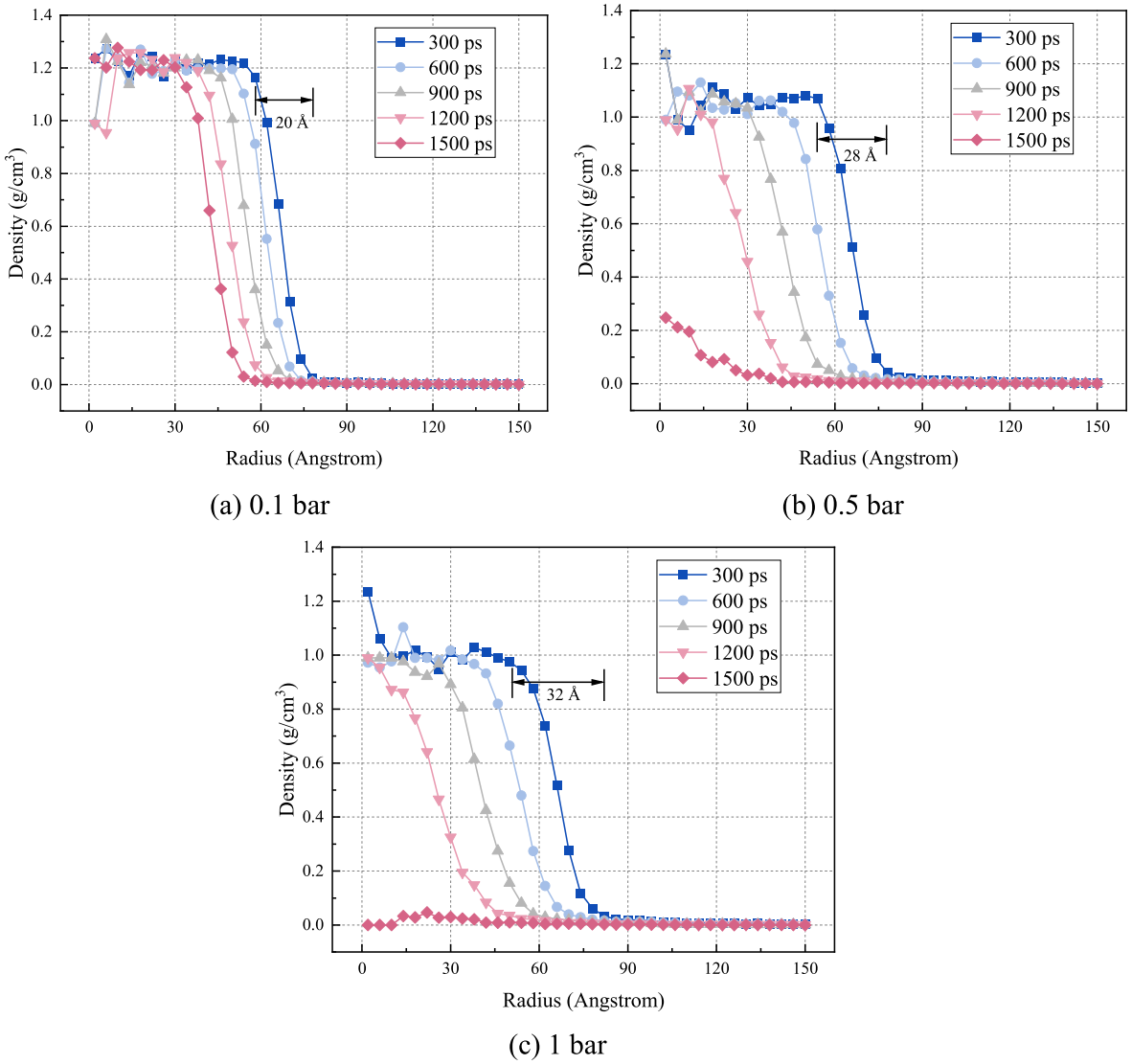


Fig. 4. Temporal radial argon density distribution under three different atmospheric pressures.

reaching its maximum at the boundary. Therefore, the intense interactions between argon and nitrogen, driven by ambient atmospheric pressure, enhance evaporation by increasing the thickness of the liquid-vapor interface. Furthermore, the greater thickness of the interface suggests that, although the initial and ambient temperatures are designed to be constant, the temperature of the liquid argon droplet varies during the evaporation process. Under higher ambient pressures, the temperature of the droplet is higher compared to conditions with lower atmospheric pressure.

3.3. Temporal argon temperature distribution

The temporal temperature of argon particles is illustrated in Fig. 5 to better understand different droplet evaporation stages. The evaporation process can be divided into three distinct stages: the heating period, the quasi-steady period, and the final evaporation period. During the heating period, argon particles are initially heated by thermostat nitrogen molecules, leading to an increase in the temperature of the argon particles. In the subsequent quasi-steady period, the temperature stabilizes, indicating that equilibrium is reached for the evaporation process. Finally, in the ending evaporation period, as the number of argon particles decreases, the remaining particles are fewer, resulting in significant fluctuations in the temperature profile.

With the thermostat set at 120 K, the heating effect on argon varies under different pressures. Under 1 bar, the heating effect results in an argon temperature of approximately 116 K between 300 ps and 900 ps, while under 0.5 bar, the temperature reaches around 113 K, and under 0.1 bar, it is around 104 K. The temperature differences between these scenarios are 3 K and 9 K, respectively. The interaction strength between nitrogen and argon is directly related to these argon temperatures. It can be concluded that, at lower nitrogen densities, the contribution to the increase in argon temperature is correspondingly reduced.

A surprising temperature drop is observed at the beginning of droplet evaporation under 0.1 bar, which is not seen in the other scenarios. This temperature drop suggests that the interaction between thermostat nitrogen molecules and argon particles is relatively weak. The primary heat required for evaporation comes from the deeper layers of the argon droplet, much like the evaporation into a vacuum [44]. This also could explain why the initial evaporation rate is significantly higher than at the other two pressures. However, as interactions with the thermostat nitrogen molecules continue, the argon droplet gradually heating up, leading to further evaporation.

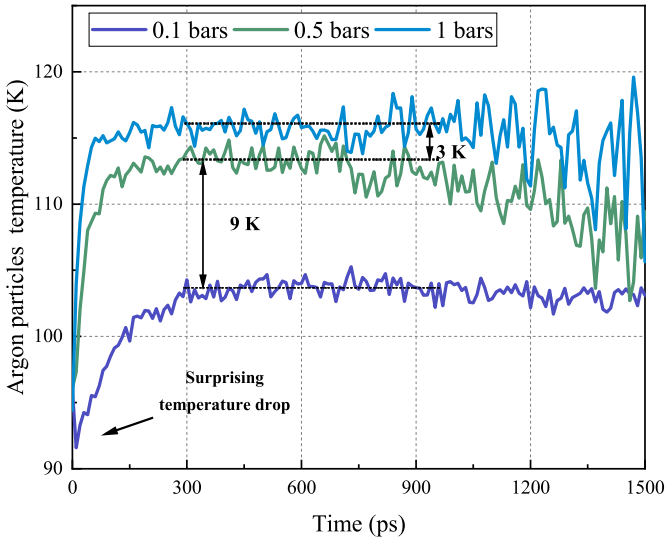


Fig. 5. Temporal temperature of remaining argon particles under different atmospheric pressures.

3.4. Comparation between with D^2 law

When the evaporation process reaches a quasi-steady state and the hydrodynamic effects can be neglected, the rate of change in the droplet's diameter can be considered constant. Thus, in this section, we focus on the time period from 200 ps to 900 ps in our simulation and compare our MD results with the traditional D^2 law. One goal of this comparison is to validate our MD results, while the other is to investigate the prediction accuracy of the D^2 law under low atmospheric pressures.

The droplet evaporation behaviors can be described by the D^2 law [49]. The law is based on the continuity assumption, and the vapor is at saturation state near the liquid-gas interface. Thus, the evaporation behavior can be simplified into a convection/diffuse process between the interface and the surrounding air. The law can be written as,

$$\frac{dD^2}{dt} = -K \tag{2}$$

where K is the evaporation constant

$$K = \frac{8\lambda_g}{\rho_l c_{p,g}} \ln\left[1 + \frac{c_{p,g}(T_\infty - T_s)}{h_g}\right] \tag{3}$$

where λ_g is the thermal conductivity of surrounding gases, ρ_l is the density of the liquid, $c_{p,g}$ is the heat capacity of gases, T_∞ is the heated temperature, T_s is the liquid surface temperature, h_g is the latent heat vaporization.

The corresponding parameters are calculated in the following Table 1. The liquid temperature is calculated as the average temperature of the argon, as shown in Fig. 5 [31], while the ambient temperature is at 120 K. As predicted by the D^2 law, the evaporation constant increases as the ambient atmospheric pressure increases.

For comparison with the D^2 law, the evaporation constant acquired by the MD results are calculated in Table 2. As indicated in Figs. 5 and 200 ps–900 ps can be regarded as quasi-steady state on both three conditions studied. The diameter of the argon droplet is defined half of the density of the bulk liquid [40]. The multiples are calculated by dividing K_{MD} , derived from the MD results, by K_{D^2}, calculated using the D^2 law.

It has been reported that the D^2 law maintains good accuracy under high atmospheric pressures, such as at 30 bar [50]. However, in their results, the evaporation constant obtained from MD simulations is approximately 7 times higher than the evaporation constant K predicted by the D^2 law at 4 bars. A good agreement is achieved with our results at 1 bar, the constant acquired by MD results is about 9 times higher than the constant predicted by D^2 law. But with the atmospheric pressure decreases, the discrepancy between two results increases dramatically.

This overprediction can be explained by the invalidity of continuum assumptions. The hydrodynamic-based models always predict a larger evaporation rate than kinetics models [51]. The increase of this overprediction with increasing of the ambient pressure can be further explained by the evaporation rate dramatically decreased in the lower atmospheric pressure conditions, as we analyzed in section 3.1. These discrepancies indicates that the accuracy of D^2 law will reduce with the decrease of the ambient atmospheric pressures.

Table 1
The evaporation constant K predicted by D^2 law.

	λ_g (W/mK)	ρ_i (kg/m3)	$c_{p,g}$ kJ/(kg/K)	$T_\infty - T$ (K)	h_g (kJ/kg)	K_{D^2} (m2/s)
1 bar	0.0113	1313.72	0.741	4	162.712	2.09×10^{-6}
0.5 bar	0.0112		0.735	7		2.88×10^{-6}
0.1 bar	0.0112		0.730	16		6.48×10^{-6}

Table 2
Comparation between evaporation constant acquired by D^2 law and MD results.

	K_{D^2}	K_{MD}	Multiple
1 bar	2.09×10^{-6}	1.79×10^{-7}	9.35
0.5 bar	2.88×10^{-6}	1.48×10^{-7}	19.50
0.1 bar	6.48×10^{-6}	8.23×10^{-8}	78.71

4. Conclusion

The heat and mass transfer characteristics of the non-equilibrium evaporation process of liquid argon were studied using the LAMMPS package. The effects of pressure are analyzed through evaporation rates, density distribution, and temperature distribution. MD results are compared with the traditional D^2 law, and the reasons for the discrepancies are discussed. The key findings are as follows.

1) The mechanism by which low atmospheric pressure affects liquid droplet evaporation can be summarized as follows: reduced interactions between the air and the liquid lead to lower heat absorption by the liquid and a thinner liquid-gas interface, resulting in a lower evaporation rate.
2) The increased initial evaporation rate at 0.1 bar can be explained by its similarity to evaporation into a vacuum. However, this elevated evaporation rate does not persist due to subsequent interactions with air particles.
3) The D^2 law is not appropriate to predict droplet evaporation behaviors under low atmospheric pressure conditions. The discrepancies between D^2 law and the MD results increases as the vacuum degree increases.

CRediT authorship contribution statement

Zhijun Tian: Writing – original draft, Methodology, Investigation. **Yanfeng Liu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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